Spectrum Typst

Feature Showcase

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Chapter 1
Theorem Environments

1.1 All Environment Types

Theorem 1.1.1 Mean Value Theorem

If f is continuous on $[a, b]$ and differentiable on (a, b), then

$$\exists c \in (a, b) : f(b) - f(a) = f'(c)(b - a).$$

Proof. Apply Rolle's theorem to $g(x) = f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x - a)$. □

Lemma 1.1.1 Triangle Inequality

For any vectors $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$:

$$|\boldsymbol{u} + \boldsymbol{v}| \leq |\boldsymbol{u}| + |\boldsymbol{v}|.$$

Definition 1.1.1 Group

A *group* $(G, *)$ is a set G with a binary operation $*$ satisfying:

1. *Closure:* $\forall a, b \in G, a * b \in G$.
2. *Associativity:* $(a * b) * c = a * (b * c)$.
3. *Identity:* $\exists e \in G, \forall a \in G : e * a = a * e = a$.
4. *Inverse:* $\forall a \in G, \exists a^{-1} \in G : a * a^{-1} = a^{-1} * a = e$.

Proposition 1.1.1 Subgroup Criterion

A non-empty subset H of a group G is a subgroup iff $ab^{-1} \in H$ for all $a, b \in H$.

Property 1.1.1 Determinant Multiplicativity

For any $n \times n$ matrices A and B:

$$\det(AB) = \det(A) \cdot \det(B).$$

Example 1.1.1 Continuous but Not Differentiable

The Weierstrass function $W(x) = \sum_{n=0}^{\infty} a^n \cos(b^n \pi x)$ is continuous everywhere but differentiable nowhere, for $0 < a < 1$ and odd b with $ab > 1 + 3\frac{\pi}{2}$.

Remark 1.1.1

While continuity is necessary for differentiability, it is not sufficient. The construction above is a classic counterexample.

Problem 1.1.1

Prove that a continuous function on a compact set is uniformly continuous.

Solution. Let $f : K \to \mathbb{R}$ be continuous with K compact. Given $\varepsilon > 0$, for each $x \in K$ choose δ_x via continuity. The open cover $\left\{B_{\frac{\delta_x}{2}}(x)\right\}$ has a finite subcover. Let $\delta = \min_i \frac{\delta_{x_i}}{2}$. For $|x - y| < \delta$, both lie in some $B_{\delta_{x_i}}(x_i)$, so $|f(x) - f(y)| < \varepsilon$. ■

Chapter 2
Tables & Equations

2.1 Striped Tables

All tables have alternating row colors automatically.

Function	Derivative	Integral	Series
e^x	e^x	$e^x + C$	$\sum_{n=0}^{\infty} \frac{x^n}{n!}$
$\sin x$	$\cos x$	$-\cos x + C$	$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$
$\cos x$	$-\sin x$	$\sin x + C$	$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$
$\ln x$	$\frac{1}{x}$	$x\ln x - x + C$	—

2.2 Labeled Equations

Equations with labels are numbered by section and can be cross-referenced.

Definition 2.2.1 Gradient and Hessian

The gradient of $f : \mathbb{R}^n \to \mathbb{R}$ is

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n} \right)^T. \tag{2.2.1}$$

The Hessian matrix is

$$\boldsymbol{H}_f = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \cdots & \cdots & \cdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}. \tag{2.2.2}$$

At a local minimum, Eq (2.2.0) vanishes and Eq (2.2.1) is positive semidefinite. See Theorem 1.1.1 for the foundation of gradient-based optimization.

Chapter 3
Pseudocode & Algorithms

3.1 Gradient Descent

Gradient Descent():

Input: objective $f : \mathbb{R}^n \to \mathbb{R}$, initial $\boldsymbol{x}_0$, step $\alpha > 0$, tolerance ε
$k := 0$
while $|\nabla f(\boldsymbol{x}_k)| > \varepsilon$ **do**
    // Compute descent direction
    $\boldsymbol{d}_k := -\nabla f(\boldsymbol{x}_k)$
    // Update with step size
    $\boldsymbol{x}_{k+1} := \boldsymbol{x}_k + \alpha \boldsymbol{d}_k$
    $k := k + 1$
    return $\boldsymbol{x}_k$

3.2 Newton's Method

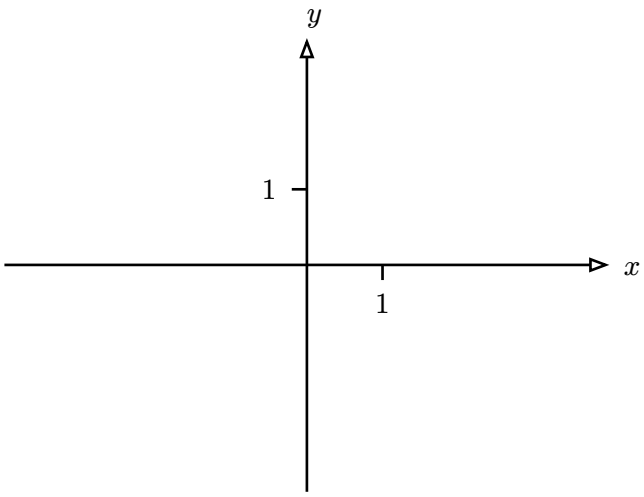
Newton's Method():

Input: f, $\boldsymbol{x}_0$, tolerance ε
$k := 0$
while $|\nabla f(\boldsymbol{x}_k)| > \varepsilon$ **do**
    // Compute Newton direction
    Solve $\boldsymbol{H}_{f(\boldsymbol{x}_k)} \boldsymbol{d}_k = -\nabla f(\boldsymbol{x}_k)$
    $\boldsymbol{x}_{k+1} := \boldsymbol{x}_k + \boldsymbol{d}_k$
    $k := k + 1$
    return $\boldsymbol{x}_k$

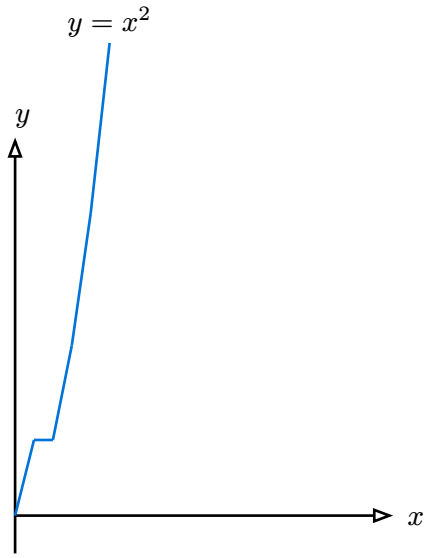
> **Remark 3.2.1**
>
> Gradient descent has linear convergence, while Newton's method has quadratic convergence near the optimum, at the cost of computing and inverting the Hessian at each step.

Chapter 4
Vector Graphics

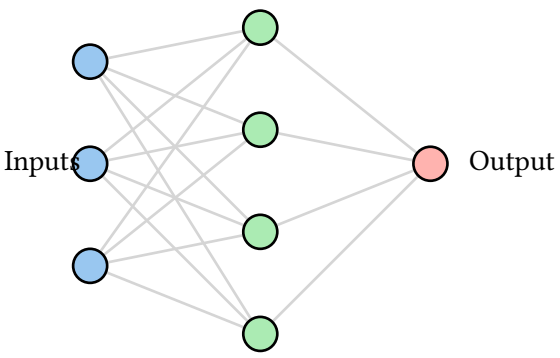
4.1 Coordinate System



4.2 Parabola Plot



4.3 Neural Network Sketch



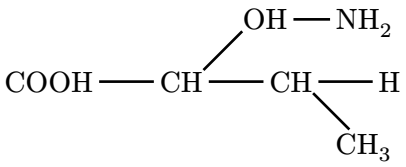
Remark 4.3.1

A feedforward neural network with one hidden layer. The universal approximation theorem guarantees that such a network can approximate any continuous function on a compact set, given enough hidden neurons.[1]

[1] C. Cybenko, *Approximation by superpositions of a sigmoidal function*, 1989.

Chapter 5
Chemistry & Physics

5.1 Chemical Structure



A generic amino acid backbone with side chains.

5.2 Reaction Thermodynamics

Reaction	Type	ΔH	ΔS
$\mathrm{N_2 + 3H_2 \rightarrow 2NH_3}$	Haber	$-92.4\,\mathrm{kJ\,mol^{-1}}$	$-198.8\,\mathrm{J^{-1}}$
$\mathrm{C + O_2 \rightarrow CO_2}$	Combustion	$-393.5\,\mathrm{kJ\,mol^{-1}}$	$2.9\,\mathrm{J^{-1}}$
$\mathrm{CaCO_3 \rightarrow CaO + CO_2}$	Decomposition	$178.3\,\mathrm{kJ\,mol^{-1}}$	$160.4\,\mathrm{J^{-1}}$

5.3 Physics Notation

Property 5.3.1 Maxwell's Equations in Vacuum

$$\nabla \cdot \boldsymbol{E} = 0, \quad \nabla \cdot \boldsymbol{B} = 0,$$

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}, \quad \nabla \times \boldsymbol{B} = \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}.$$

Theorem 5.3.1 Electromagnetic Wave Equation

Taking the curl of Faraday's law and substituting Ampère's law yields the wave equation:

$$\nabla^2 \boldsymbol{E} = \mu_0 \varepsilon_0 \frac{\partial^2 \boldsymbol{E}}{\partial t^2},$$

describing waves propagating at $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 2.998 \cdot 10^{8}\,\mathrm{m\,s^{-1}}$.

Example 5.3.1 Visible Light

Visible wavelengths range from $(380 - 750)\,\mathrm{nm}$, corresponding to frequencies of $(400 - 790)\,\mathrm{THz}$. The energy per photon is $E = h\nu \approx (1.65 - 3.27)\,\mathrm{eV}$.

Remark 5.3.1 Unit Formatting

All physical quantities above use the `unify` package: `qty` for values with units and `qtyrange` for ranges. Unit formatting is automatic and follows SI conventions.

Chapter 6
中文示例

6.1 定积分与极限

定理 6.1.1 牛顿-莱布尼茨公式

若 F 是 f 在 $[a,b]$ 上的一个原函数, 则

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

证明. 将 $[a,b]$ 分成 n 等份, 利用拉格朗日中值定理. □

定义 6.1.1 一致连续

设 $f: I \to \mathbb{R}$. 若对任意 $\varepsilon > 0$, 存在 $\delta > 0$, 使得对任意 $x, y \in I$, 当 $|x-y| < \delta$ 时恒有

$$|f(x) - f(y)| < \varepsilon,$$

则称 f 在 I 上一致连续.

问题 6.1.1

证明 $f(x) = \sqrt{x}$ 在 $[0,\infty)$ 上不一致连续, 但在 $[1,\infty)$ 上一致连续.

解. 取 $x_n = \frac{1}{n^2}$, $y_n = \frac{1}{4n^2}$, 则 $|x_n - y_n| \to 0$ 但 $|f(x_n) - f(y_n)| = \frac{1}{2n}\neg \to 0$, 故不一致连续. 在 $[1,\infty)$ 上, $|f(x)-f(y)| = |x-y\frac{|}{\sqrt{x}+\sqrt{y}} \le |x-y\frac{|}{2}$, 满足 Lipschitz 条件.

6.2 物理常数

常数	符号	数值
真空光速	c	$2.998 \cdot 10^{8}\ \mathrm{m\,s^{-1}}$
普朗克常数	h	$6.626 \cdot 10^{-34}\ \mathrm{J\,s}$
万有引力常数	G	$6.674 \cdot 10^{-11}\ \mathrm{m^{3}\,kg^{-1}\,s^{-2}}$
阿伏伽德罗常数	N_A	$6.022 \cdot 10^{23}\ \mathrm{mol^{-1}}$

注解 6.2.1

定理 1.1.1 可推广至多维: 若 $f: \mathbb{R}^n \to \mathbb{R}$ 可微, 则存在 ξ 使得

$$f(\boldsymbol{y}) - f(\boldsymbol{x}) = \nabla f(\xi)\cdot(\boldsymbol{y}-\boldsymbol{x})$$

. 参考式 式 (2.2.0) 和 式 (2.2.1).